Almost-Idempotent Stochastic Maps — internal lab-book

(scientific-exploration repo; see PRD.md / CLAUDE.md)

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1 Roadmap and rigour status

This lab-book tracks the report-facing slice of the registry in `argument/`. The north star is `op-classical`; its status is *open*, and the status-ledger anchor is 1. The definitions used by the registry are the canonical shards `def-stochastic`, `def-almost-idempotent`, `def-near-positive-projection`, `def-signed-idempotent`, `def-negative-mass`, `def-height`, `def-visible-set`, `def-invisible-mass`, and `def-exposed`. This report references those identifiers; it does not introduce parallel definitions.

The rigorous line is deliberately narrow. A result may be called rigorous here only when it is byte-matched to a local theorem under `refs/`, validated in an `af` workspace, or Lean-proved. At the time of this shard, the only registry results reproduced as rigorous are the five `af`-validated entries 1, 2, 3, 4, and 5. Every other registry result remains anchored only in the status ledger, with its lower-rung status left intact.

The working headline is also status-sensitive. The inherited campaign distinguishes the realizable-family linear law from the worst-case sqrt-envelope: the linear relation is the mechanism seen in the reduction record, while the sqrt scale is the stability target. The sharp-exponent obstruction is `ex-hume`, anchored at 1, but it is still `proved-mod-audit` here rather than one of the rigorous results. The bridge between the signed and stochastic pictures is the validated result 1; crossing between those pictures should cite that registry id rather than re-normalising notation locally.

2 The signed-stochastic bridge

Lemma 1 (Registry contract for `lem-classical-equiv`). *The signed-idempotent and stochastic-idempotent formulations of classical stability are equivalent up to universal constants: Q row-stochastic with ||Q^2-Q|| <= eta gives P=theta(2Q-1) signed affine retraction with ||P-Q|| <= C eta and neg mass delta <= C eta, and conversely row-normalising p_i^+ gives Q with ||P-Q|| <= 2 delta, ||Q^2-Q|| <= 6 delta+4 delta^2.*

Status. `proved`; `af: validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records a 29-node adversarial tree with root `validated` and clean taint. The human-readable export is `proofs/lem-classical-equiv/export.md`; the export also records the earlier missing-definition challenge and its repair by importing `def-negative-mass`.
Role in the chain. This bridge is the authorized way to move between `op-classical`'s stochastic formulation and the signed exact-idempotent registry. It is a dependency for the inherited simplex, rank-one, and approximate-simplex reductions, but those downstream reductions remain `proved-mod-audit` unless and until they clear an L0 gate.

3 Height collapse

Proposition 2 (Registry contract for `obs-height-collapse`). *`Height collapse: for an exact signed idempotent P with 0 < delta(P) <= 1/4, nonempty visible set W(P), and hidden top vertex v (height H = dist_1(p_v, conv W) maximal among rows), the invisible mass sigma_v and the row negative mass nu_v satisfy H * (1 - sigma_v) <= nu_v * (2 + 4*delta).`*

Status. `proved`; `af: validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records a 19-node adversarial tree with root `validated` and taint `19/19 clean`. The export is `proofs/obs-height-collapse/export.md`. Earlier compound "hence" clauses were excluded from the contract before validation; this section records only the validated inequality.
Role in the chain. This result explains why invisible mass alone does not finish the Kernel/`(EX)` route: height is controlled only after pricing the complementary mass. A future dimension-free cap on the relevant invisible mass would combine with this inequality, but that cap is not proved here.

4 Mass split bookkeeping

Lemma 3 (Registry contract for `lem-mass-split`). *Mass split: for an exact signed idempotent P and any row index v, writing a_j = P_{vj}, a_j^+ = max(a_j, 0), a_j^- = max(-a_j, 0), and nu_v = sum_j a_j^-, one has sum_j a_j^+ = 1 + nu_v.*

Status. `proved`; `af: validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records a clean 9-node adversarial tree with root `validated` and taint `9/9 clean`. The export is `proofs/lem-mass-split/export.md`.
Role in the chain. This is the mass bookkeeping dependency factored out of the first `conj-halo-collapse` elevation attempt. Its purpose is to keep later halo-collapse proof trees from re-deriving the same row-sum identity in multiple sibling branches.

5 Convex outsourcing

Lemma 4 (Registry contract for `lem-residual-lower`). *Convex outsourcing: let C be the convex hull of finitely many points of R^n, and suppose p = sum_{j=1}^m c_j p_j + (1 - s) q with points p_j, q in R^n, coefficients c_j >= 0, s = sum_{j=1}^m c_j < 1, and dist_1(p_j, C) <= dist_1(p, C) for every j; then dist_1(p, C) <= dist_1(q, C).*

Status. `proved`; `af: validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records 32 validated live nodes, 36 total nodes including 4 archived nodes, root `validated`, and taint `36/36 clean`. The export is `proofs/lem-residual-lower/export.md`.
Role in the chain. This is a frame-free convex geometry dependency for `conj-halo-collapse`. It supplies the lower residual-distance comparison used when a hidden row is split into ordinary mass plus a residual term.

6 Residual distance bound

Lemma 5 (Registry contract for `lem-residual-upper`). *Residual distance bound: let C be the convex hull of finitely many points of R^n, let b_1..b_M, c_1..c_N >= 0 with m = sum_j b_j - sum_k c_k > 0, let p_j, r_k be points of R^n with q = (sum_j b_j p_j - sum_k c_k r_k) / m, and let D_k >= 0 satisfy ||x - r_k||_1 <= D_k for all x in C and each k; then m * dist_1(q, C) <= sum_j b_j * dist_1(p_j, C) + sum_k c_k * D_k.*

Status. `proved`; `af: validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records 49 validated live nodes, 52 total nodes including 3 archived nodes, root `validated`, and taint `52/52 clean`. The export is `proofs/lem-residual-upper/export.md`. The tree size is above the 12-node brittleness target, so the tracked refactor warning remains honest rather than being hidden in prose.
Role in the chain. This is the companion frame-free convex geometry dependency for `conj-halo-collapse`. Together with 3 and 4, it packages the residual-recombination estimate needed by the halo-collapse bridge, which remains a conjectural node until validated separately.

7 Status ledger for non-validated registry results

The rows below are anchors, not upgrades. They give every remaining registry result a report label while preserving the registry status. The five `af`-validated results are reproduced in 1–5; they are not duplicated in this ledger.

Table 1: Registry status ledger for report anchors outside the five validated sections.

Registry id	Status	Contract gloss
`conj-ex`	conjecture	`(EX)` chart-bound working form for exact signed idempotents; still a conjectural input equivalent to `conj-kernel` in the registry.
`conj-halo-collapse`	conjecture	Halo-robust height-collapse bridge importing `lem-mass-split`, `lem-residual-lower`, and `lem-residual-upper`; not yet validated.
`conj-kernel`	conjecture	Kernel Conjecture, the geometric input intended to close `op-exposed-hull` and hence `op-classical`.
`conj-no-free-frontier`	conjecture	Exposedness-absorption statement from the frontier analysis; retained as conjectural and not elevated.
`ex-hume`	proved-mod-audit	Explicit 3-by-3 sharpness family for the exponent in `op-classical`; inherited proof is not rigorous here.
`lem-canonical-separator`	proved-mod-audit	Canonical separator package for a hidden top vertex and its harmonic deficit; inherited/mod-audit only.
`lem-dual-localization`	open	Frame-free dual-localization gap behind the linear-law route; explicitly open.
`lem-exposed-circuit`	proved-mod-audit	Concentration and circuit estimate for exposed vertices; inherited/mod-audit only.
`lem-factorization`	proved-mod-audit	Factorization bound connecting chart scores to the `(EX)` input; constants retained as mod-audit.
`lem-leakage`	proved-mod-audit	Affine-face leakage estimate for almost-idempotent stochastic rows; inherited/mod-audit only.
`lem-wiggle-rigidity`	proved-mod-audit	F-WR wiggle-rigidity estimate for common-pattern webs; side conditions remain inherited.
`obs-deep-leakage`	heuristic	Heuristic obstruction showing why a purely local row-identity argument cannot price deep-side mass.
`obs-fwr-gap`	heuristic	F-WR forbidden-gap obstruction to using web rigidity as the dimension-free shallow-class counter.
`obs-linear-law-finite-delta`	numerical	Exact finite-delta certificate exceeding the empirical linear-law constant; L3 evidence, not proof.
`obs-sigma-halo-nonrobust`	numerical	Exact certificate that the literal zero-halo sigma cap is false; L3 evidence and obstruction.
`op-classical`	open	North-star classical projection stability problem; still open in this repo.
`op-exposed-hull`	open	Global exposed-hull lemma that would imply `op-classical` through the inherited factorization route; open.
`prop-approx-simplex`	proved-mod-audit	Approximate-simplex reduction to a stochastic idempotent; inherited/mod-audit only.
`thm-classical-factorization`	proved-mod-audit	Conditional classical factorization theorem through cluster geometry and `op-exposed-hull`; not rigorous here.

`thm-cluster`	proved-mod-audit	Cluster geometry reduction for signed affine retractions; inherited/mod-audit only.
`thm-corner-constants`	proved-mod-audit	Corner-scale constants, with numerical components explicitly not promoted.
`thm-rank-one`	proved-mod-audit	Rank-one signed affine retraction stability, including the Hume family; inherited/mod-audit only.
`thm-simplex`	proved-mod-audit	Simplex and one-dimensional row-polytope stability reduction; inherited/mod-audit only.
`thm-well-exposed`	proved-mod-audit	Well-exposed-vertex route into the simplex theorem; inherited/mod-audit only.